Pavel Vasilyev

Senior AI Engineer - Agentic Systems

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PROFILE

Senior AI engineer with fifteen years across applied ML, NLP, and software. I build LLM agent infrastructure and measure it: multi-provider routing, enforced boundaries, and eval harnesses that report pass rate, cost, and latency for every change.

FIT FOR THE ROLE

- Evaluation first: a probe-suite harness reporting pass rate, cost, and wall time per run, used to A/B agent configuration changes
- Model-agnostic: Claude, Gemini, OpenRouter, Ollama, and Cloudflare Workers AI behind one router with role-specific fallback and token and latency tracing
- Enforced boundaries, not prompt rules: pre-commit and scope-guard hooks around agents; a credential service with atomic replacement
- Python and function-calling tools in a fallback-aware harness, with unit tests for backoff and model selection

EXPERIENCE

Independent Software Practice - consulting and open source 2010-present

Consulting, advisory, and software practice between salaried roles.

- Built tools and technical systems against client specifications while maintaining an open-source body of 150+ projects across eight or more languages.

Research Lead - Spring Research 2020-2022

Quantitative research firm focused on financial-market modelling.

- Owned the full applied-research cycle: monitored financial markets, analysed data, developed new signal representations and transformations, extracted ML features, selected approaches, and built and trained models.

Earlier Data Science, NLP & Software Roles - Nestlogic, Maverick Medical AI, Athena Portfolio Solutions, Ten-X, Verba, and Finfon 2009-2019

Roles in Tel Aviv, Hong Kong, and Prague.

- Financial-news NLP and signal extraction; medical-text classification and information extraction; computer-vision pipelines; and corpus-linguistics infrastructure.

SELECTED WORK (PRIVATE REPOSITORIES, CODE ON REQUEST)

pi-harness - Eval harness for a coding agent, scoring pass/fail, cost, and wall time per run. A/B of a structural gate: spec-first 0/3 to 3/3 at +26% cost and +80% wall time (n=3, one model).

inference_harness - Elixir LLM harness for multi-provider routing, role-specific fallback, token/latency tracing, and bounded concurrent jobs.

free-llm-harness - Python harness running prompt sets against free OpenRouter models, with backoff, model selection, and tool calls. Over n=88 requests: 100% success on the quota probe, 83% in the outage probe, p95 6.4 s.

BACKGROUND

Education: MSc Computational Linguistics, RSUH, Moscow, 2015. BSc Nuclear Physics, Czech Technical University, Prague, 2009.

Languages: English, Hebrew, Russian; working knowledge of German and French.